

Sprint 2: Active Ability Rings — Detailed Implementation Plan

Project: D&D 3.5e Unity Prototype
Sprint: 2 of 4 (Ring Implementation)
Scope: 9 Active Ability Rings with Command Word Activation
Rules Authority: Core D&D 3.5e ONLY (PHB, DMG, MM)
Date: May 2026
Prerequisites: Sprint 1 complete — 36 passive ring variants, RingFactory, RingDatabase, RingNames infrastructure

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1. Executive Summary

Sprint 2 adds **9 rings with active abilities** that require player interaction — command word activation, use-frequency tracking, charge systems, and special mechanics. These rings build on Sprint 1's passive ring infrastructure (RingFactory, RingDatabase, RingNames, ItemData ring fields) and leverage numerous existing systems (spell casting, summoning, invisibility, blink, spell turning).

Key Deliverables

- **Command Word Activation System** — New `TryUseRing()` method following the `TryUseStaff()` pattern
- **Use Tracking Systems** — Daily uses (3/day), weekly uses (1/week, 3/week), charge-based (50 charges, regen 10/day)
- **Ring Ability Selection Panel** — For multi-ability rings (Ring of Shooting Stars)
- **5 rings leveraging existing spell effects** — Invisibility, Blink, Telekinesis, Spell Turning, summoning
- **4 rings requiring new effect implementations** — Animal Friendship (Charm Animal), Ram (force bolt + bull rush), X-Ray Vision, Shooting Stars (5 unique abilities)
- **1 new creature definition** — Noble Djinni for Ring of Djinni Calling

Complexity Breakdown

Tier	Rings	Est. Effort
Simple (reuse existing spell)	Invisibility, Blink, Telekinesis	2-3 days
Moderate (new effect + tracking)	Animal Friendship, Ram, Spell Turning	4-5 days
Complex (multi-ability or new systems)	X-Ray Vision, Shooting Stars, Djinni Calling	6-8 days
Infrastructure	Activation system, UI, tracking	3-4 days
Total Estimate		15-20 days

2. Sprint 1 Foundation Recap

Existing Ring Infrastructure (commit `a77a6f7`)

File	Purpose	Key Elements
<code>RingNames.cs</code>	String constants for all ring names	<code>RING_OF_INVISIBILITY</code> , <code>RING_OF_BLINKING</code> , etc.
<code>RingFactory.cs</code>	Creates ring <code>ItemData</code> instances	<code>CreateRing(ringName)</code> with stat bonuses, metadata
<code>RingDatabase.cs</code>	Registry of all ring definitions	<code>Initialize()</code> , <code>Get(name)</code> , ring lookup
<code>ItemData.cs</code>	Ring fields on item model	<code>IsRing</code> , <code>RingSlot</code> , ring-specific properties
<code>CharacterStats.cs</code>	Stat modification from rings	Ring bonus application on equip/unequip
<code>Inventory.cs</code>	Ring slot management	Two ring slots, equip/unequip handling
<code>ItemDatabase.cs</code>	Master item registry	Ring entries integrated

What Sprint 1 Does NOT Have (Sprint 2 Must Add)

- No `TryUseRing()` method — rings cannot be “used” as an action
- No ring activation UI — no way to trigger command word abilities
- No daily/weekly/charge tracking on rings

- No charge regeneration on rest
- No multi-ability ring selection panel
- All 9 Sprint 2 rings exist in RingDatabase as **passive-only stubs** (name/price/slot but no active effects)

3. Ring Specifications — All 9 Rings

3.1 Ring of Invisibility

Property	Value
Name Constant	RingNames.RING_OF_INVISIBILITY
Market Price	20,000 gp
Activation	Command word (standard action)
Frequency	At will (unlimited)
Effect	Casts Invisibility (PHB p.245) on the wearer
Duration	Per spell: 1 min/level (10 rounds at CL 3, but ring CL = 3 per DMG)
Caster Level	3rd
Special Rules	Breaks on attack or offensive action per standard Invisibility rules
Aura	Faint illusion

Implementation Notes:

- Calls existing `CharacterController.ApplyInvisibilityEffect()` (found at `CharacterController.cs:1708`)
- `HasActiveInvisibilityEffect` property already tracks active state (`CharacterController.cs:1535`)
- Duration: 30 rounds (CL 3 × 10 rounds/level... but Invisibility is 1 min/level = 10 rounds per CL)
- Actually per DMG, ring CL = 3, so duration = 3 minutes = 30 rounds
- No new systems needed beyond the activation framework

Pseudocode:

```

TryUseRing_Invisibility(caster, ring):
    if not CanUseItemManipulationAction(caster, ActionType.Standard):
        return false
    durationRounds = 30 // CL 3 × 10 rounds/level
    caster.CharacterController.ApplyInvisibilityEffect(durationRounds, caster)
    ConsumeItemManipulationAction(caster)
    LogMessage("{caster.Name} activates Ring of Invisibility")
    return true

```

3.2 Ring of Blinking

Property	Value
Name Constant	RingNames.RING_OF_BLINKING
Market Price	27,000 gp
Activation	Command word (standard action)
Frequency	At will (unlimited)
Effect	Casts Blink (PHB p.206) on the wearer
Duration	Per spell: 1 round/level = 7 rounds (CL 7 per DMG)
Caster Level	7th
Special Rules	50% miss chance on incoming attacks; 20% miss chance on wearer's attacks; can move through solid objects
Aura	Moderate transmutation

Implementation Notes:

- HasActiveBlinkEffect property exists (CharacterController.cs:1583)
- Full blink miss chance system is already implemented in combat resolution
- Duration: 7 rounds (CL 7)
- Straightforward activation — same pattern as Invisibility

Pseudocode:

```
TryUseRing_Blink(caster, ring):  
    if not CanUseItemManipulationAction(caster, ActionType.Standard):  
        return false  
    durationRounds = 7 // CL 7 × 1 round/level  
    caster.CharacterController.ApplyBlinkEffect(durationRounds, caster)  
    ConsumeItemManipulationAction(caster)  
    LogMessage("{caster.Name} activates Ring of Blinking")  
    return true
```

3.3 Ring of Animal Friendship

Property	Value
Name Constant	RingNames.RING_OF_ANIMAL_FRIENDSHIP
Market Price	10,800 gp
Activation	Command word (standard action)
Frequency	3/day
Effect	Casts Charm Animal (PHB p.208) on target animal
Duration	1 hour/level = 1 hour (CL 1... but DMG says functions as charm animal)
Caster Level	Not specified in DMG entry — use minimum CL 1 for druid spell
Save	Will DC 11 negates
Target	One animal within 25 ft (close range at CL 1)
Special Rules	Only works on Animal type creatures; can have up to 12 HD of charmed animals simultaneously (as per the ring description in DMG)
Aura	Faint enchantment

Implementation Notes:

- **Charm Animal spell is NOT currently implemented** — only exists as a stub name in StaffDatabase.cs:538
- Needs new spell effect: simplified charm that changes creature disposition/team
- Must validate target is Animal type (check creature type field)
- **Daily use tracking** — first ring requiring the daily uses system
- HD cap: Sum of charmed animal HD $\leq$ 12 (ring-specific rule from DMG p.232)

New Systems Required:

1. Charm Animal effect (new spell implementation)
2. Daily use counter on ring items
3. Target validation (animal type only)
4. Charmed animal HD tracking per ring

Pseudocode:

```

TryUseRing_AnimalFriendship(caster, ring):
    if not CanUseItemManipulationAction(caster, ActionType.Standard):
        return false
    if ring.DailyUsesRemaining <= 0:
        LogMessage("Ring of Animal Friendship has no uses remaining today")
        return false
    // Enter targeting mode
    StartTargetingMode(caster, TargetType.SingleEnemy, range: 25ft, callback: (target)
=>
        if target.CreatureType != CreatureType.Animal:
            LogMessage("Target must be an animal")
            return
        if GetCharmedAnimalHDTototal(caster, ring) + target.HitDice > 12:
            LogMessage("Cannot charm: would exceed 12 HD limit")
            return
        willSaveDC = 11
        if target.RollWillSave() >= willSaveDC:
            LogMessage("{target.Name} resists the charm")
        else:
            ApplyCharmAnimalEffect(target, caster, durationRounds: 600) // 1 hour
            LogMessage("{target.Name} is charmed by {caster.Name}")
        ring.DailyUsesRemaining -= 1
        ConsumeItemManipulationAction(caster)
    )
    return true

```

3.4 Ring of the Ram

Property	Value
Name Constant	<code>RingNames.RING_OF_THE_RAM</code>
Market Price	8,600 gp
Activation	Command word (standard action)
Frequency	Charge-based: 50 charges max, expend 1–3 per use, regenerates 10 charges/dawn (see DMG errata — not “1d10” but a flat 10 from the 2003 errata context; however DMG p.233 says “regain 1d10 charges per day”)
Effect	Force bolt deals damage and/or bull rush
Damage	1d6 per charge expended (1 charge = 1d6, 2 charges = 2d6, 3 charges = 3d6)
Range	Close (25 ft + 5 ft/2 levels = 50 ft at CL 9)
Bull Rush	Strength 25 (+7 modifier) check; +1 per charge expended
Caster Level	9th
Attack Roll	Ranged touch attack using wearer’s stats
Special	Can target objects; deals double damage to objects
Aura	Moderate evocation

Implementation Notes:

- **New charge system** with regeneration — first ring using charges
- Charges tracked via `ItemData.CurrentCharges` / `ItemData.MaxCharges` (may need to add these fields)
- Regeneration: Add charge restoration logic to rest handler (`GameManager.cs:935-980`)
- Requires ranged touch attack roll (existing combat system supports touch attacks)
- Bull rush uses existing bull rush mechanics if available, or simplified version
- Player chooses 1–3 charges to expend → needs a small UI for charge selection

New Systems Required:

1. Charge tracking on ItemData (CurrentCharges, MaxCharges)
2. Charge regeneration in rest handler
3. Charge selection UI (1/2/3 charges)
4. Ranged touch attack from ring
5. Force bolt damage + bull rush effect

Pseudocode:

```

TryUseRing_Ram(caster, ring):
  if not CanUseItemManipulationAction(caster, ActionType.Standard):
    return false
  if ring.CurrentCharges <= 0:
    LogMessage("Ring of the Ram has no charges remaining")
    return false
  // Open charge selection panel (1, 2, or 3)
  maxCharges = Min(3, ring.CurrentCharges)
  OpenChargeSelectionPanel(maxCharges, callback: (chargesSpent) =>
    StartTargetingMode(caster, TargetType.SingleEnemy, range: 50ft, callback:
(target) =>
  // Ranged touch attack
  attackRoll = RollRangedTouchAttack(caster, target)
  if attackRoll.IsHit:
    // Damage
    damage = Roll(chargesSpent, d6)
    if target.IsObject:
      damage *= 2
    target.TakeDamage(damage, DamageType.Force)
    // Bull rush
    bullRushStr = 25 + chargesSpent // +1 per charge
    bullRushCheck = Roll(1, d20) + GetStrMod(bullRushStr)
    targetCheck = Roll(1, d20) + target.GetStrMod() + target.SizeModifier
    if bullRushCheck > targetCheck:
      PushTarget(target, distance: 5ft * (bullRushCheck - targetCheck))
  ring.CurrentCharges -= chargesSpent
  ConsumeItemManipulationAction(caster)
)
)
return true

// In rest handler:
OnRest():
  foreach ring in equippedRings where ring.IsChargeRing:
    ring.CurrentCharges = Min(ring.MaxCharges, ring.CurrentCharges + Roll(1, d10))

```

3.5 Ring of Telekinesis

Property	Value
Name Constant	RingNames.RING_OF_TELEKINESIS
Market Price	75,000 gp
Activation	Command word (standard action)
Frequency	At will (unlimited)
Effect	Casts Telekinesis (PHB p.292) — all three modes available
Modes	Sustained Force (move 375 lbs, concentration), Combat Maneuver (bull rush/disarm/grapple/trip at CL vs target), Violent Thrust (hurl objects/creatures, 375 lb limit, 15d6 max damage)
Caster Level	9th
Duration	Concentration (up to 9 rounds) or instantaneous (violent thrust)
Range	Long (400 ft + 40 ft/level = 760 ft at CL 9)
Aura	Moderate transmutation

Implementation Notes:

- Telekinesis spell constant exists: SpellNames.TELEKINESIS
- The full Telekinesis spell implementation needs to be verified for all three modes
- For the prototype, may simplify to **Violent Thrust** mode only (most combat-relevant)
- Alternatively, open a mode selection panel (Sustained Force / Combat Maneuver / Violent Thrust)
- At will, so no use tracking needed — just activation system

Prototype Simplification Recommendation:

- Implement **Violent Thrust** as primary mode: ranged attack, deals damage based on object weight
- **Combat Maneuver** as secondary: opposed CL check vs target for bull rush/trip
- **Sustained Force** as flavor only (log message, no mechanical implementation)
- This follows the same simplification approach used elsewhere in the prototype

Pseudocode:

```

TryUseRing_Telekinesis(caster, ring):
  if not CanUseItemManipulationAction(caster, ActionType.Standard):
    return false
  // Open mode selection: Violent Thrust, Combat Maneuver, Sustained Force
  OpenTelekinesisPanel(callback: (mode) =>
    switch mode:
      case ViolentThrust:
        StartTargetingMode(caster, range: 760ft, callback: (target) =>
          // Hurl objects at target
          attackRoll = RollRangedAttack(caster, target)
          if attackRoll.IsHit:
            damage = Roll(min(15, objectCount), d6)
            target.TakeDamage(damage, DamageType.Bludgeoning)
            ConsumeItemManipulationAction(caster)
          )
        case CombatManeuver:
          StartTargetingMode(caster, range: 760ft, callback: (target) =>
            // Opposed check: CL 9 + d20 vs target CMB defense
            ApplyTelekineticManeuver(caster, target, CL: 9)
            ConsumeItemManipulationAction(caster)
          )
        case SustainedForce:
          LogMessage("{caster.Name} telekinetically manipulates objects")
          ConsumeItemManipulationAction(caster)
    )
  return true

```

3.6 Ring of X-Ray Vision

Property	Value
Name Constant	RingNames.RING_OF_XRAY_VISION
Market Price	25,000 gp
Activation	Command word (standard action)
Frequency	At will, but see fatigue rules
Effect	See through solid matter up to 20 ft
Penetration	20 ft stone/wood/dirt, 10 ft iron, 1 ft gold/lead/platinum block it entirely
Scanning	10 ft × 10 ft area per round
Duration	1 minute per use (10 rounds)
Fatigue	Each use after the first in a single hour causes Con damage (1d4 Con drain) — per DMG errata the penalty is not drain but 1 point of Con damage per use after first
Caster Level	Not specified — treat as CL 5
Aura	Moderate divination

Implementation Notes:

- **No existing vision-through-walls system** — this is entirely new
- For the prototype, X-Ray Vision is primarily a **narrative/exploration tool** with limited combat application
- **Recommended prototype implementation:**
 - Apply a status effect “X-Ray Vision” for 10 rounds
 - Mechanically: reveal hidden/invisible creatures within 20 ft (grants See Invisibility-like effect)
 - Reveal secret doors / traps within 20 ft (if such system exists)
 - The “see through walls” aspect is hard to represent in a tactical combat prototype
 - Con damage on repeated use within same hour provides a meaningful cost
 - Track “uses this hour” — simpler approach: track uses since last rest, apply Con damage after first use

Prototype Simplification:

- **Combat mode:** Grants See Invisibility + reveals hidden creatures within 20 ft for 10 rounds
- **Exploration mode:** Log message indicating what is found (script-driven, encounter-specific)
- **Fatigue:** Apply 1 Con damage on second+ use per rest period (simplified from “per hour”)

Pseudocode:

```

TryUseRing_XRayVision(caster, ring):
    if not CanUseItemManipulationAction(caster, ActionType.Standard):
        return false
    // Con damage on repeated use
    ring.XRayUsesThisRest += 1
    if ring.XRayUsesThisRest > 1:
        conDamage = 1 // simplified from 1d4
        caster.ApplyAbilityDamage(AbilityScore.Constitution, conDamage)
        LogMessage("{caster.Name} strains from X-Ray vision use (1 Con damage)")
    // Apply effect
    ApplyStatusEffect(caster, "XRayVision", durationRounds: 10, effects:
        - SeeInvisibility: true
        - RevealHidden: true
        - Range: 20ft
    )
    ConsumeItemManipulationAction(caster)
    LogMessage("{caster.Name} activates X-Ray vision, seeing through barriers")
    return true

```

3.7 Ring of Shooting Stars

Property	Value
Name Constant	RingNames.RING_OF_SHOOTING_STARS
Market Price	50,000 gp
Activation	Command word (varies per ability)
Caster Level	12th
Aura	Strong evocation

Five Abilities:

3.7a Dancing Lights (at will, outdoors at night only)

Property	Value
Frequency	At will
Effect	As Dancing Lights spell (PHB p.216)
Duration	1 minute (10 rounds)
Restriction	Outdoors at night only
Action	Standard action

3.7b Light (at will)

Property	Value
Frequency	At will
Effect	As Light spell (PHB p.248)
Duration	120 minutes (CL 12)
Action	Standard action

3.7c Ball Lightning (1/day, special)

Property	Value
Frequency	1/day
Effect	Create 1-4 balls of lightning; each can be directed (move action) to strike a target
Damage	Each ball deals electricity damage on contact (per DMG: varies by number created — 1 ball = 4d6, 2 balls = 3d6 each, 3 balls = 2d6 each, 4 balls = 1d6+1 each)
Range	120 ft
Duration	Balls persist for 4 rounds or until discharged
Save	Reflex DC 13 half
Action	Standard action to create, move action to direct

3.7d Shooting Stars (3/week, outdoors at night only)

Property	Value
Frequency	3/week
Effect	Fire 1–3 shooting stars (flame strikes from above)
Damage	Each star: 12d6 fire in 5-ft radius, Reflex DC 13 half
Range	70 ft
Restriction	Outdoors at night only
Action	Standard action

Note: Per DMG p.233, the actual shooting stars ability deals less damage. Cross-referencing: each star creates a 5-ft-radius burst dealing a total of 12 points of damage (not 12d6). Let me re-verify: DMG p. 233 states “Shooting Stars: This ability functions outdoors at night... The wearer can fire up to three shooting stars... Each shooting star deals 12 points of fire damage.” So **12 flat fire damage per star, Reflex DC 13 half, 5-ft radius.**

Corrected:

| **Damage** | Each star: 12 fire damage in 5-ft radius, Reflex DC 13 half |

3.7e Faerie Fire (2/day)

Property	Value
Frequency	2/day
Effect	As Faerie Fire spell (PHB p.229)
Duration	12 minutes (CL 12)
Area	5-ft-radius burst per creature/object targeted
Effect	Outlined creatures take -20 to Hide, visible if invisible
Action	Standard action

Implementation Notes:

- **Most complex ring** — five abilities with mixed frequency tracking
- Requires **Ring Ability Selection Panel** (like StaffSpellSelectionPanel) — shows 5 abilities with remaining uses
- `DANCING_LIGHTS` spell constant exists in SpellNames
- `FAERIE_FIRE` needs to be checked / may need implementation
- Ball Lightning is entirely new

- Shooting Stars is entirely new
- Daily tracking: Ball Lightning (1/day), Faerie Fire (2/day)
- Weekly tracking: Shooting Stars (3/week)
- At-will: Dancing Lights, Light

Use Tracking Structure:

```
ring.DailyUses["BallLightning"] = { Used: 0, Max: 1 }
ring.DailyUses["FaerieFire"] = { Used: 0, Max: 2 }
ring.WeeklyUses["ShootingStars"] = { Used: 0, Max: 3 }
// Dancing Lights and Light: unlimited, no tracking
```

3.8 Ring of Spell Turning

Property	Value
Name Constant	RingNames.RING_OF_SPELL_TURNING
Market Price	98,280 gp
Activation	Automatic (no action required — always active while worn)
Frequency	Pool-based: 1d4+6 spell levels absorbed, then inert until recharged
Effect	As Spell Turning spell (PHB p.283) — reflects targeted spells back at caster
Recharge	New pool when re-equipped or on specific trigger (see notes)
Caster Level	13th
Special Rules	Only reflects spells that specifically target the wearer (not area effects); partial reflection if remaining levels < spell level; both casters affected by the same spell if partial
Aura	Strong abjuration

Implementation Notes:

- **Spell Turning is already fully implemented** at `GameManager_Spells_Phase2.cs:392-426`
- Uses `StatusEffect.CustomTag = $"SpellTurning:{turningLevels}"` to track remaining pool
- Existing implementation: self-only, rolls 1d4+6, stores remaining levels
- **Ring difference from spell:** The ring applies this effect automatically when equipped, not as a cast spell
- Ring should apply the Spell Turning status effect with 1d4+6 levels on equip

- When pool is depleted, ring becomes inert (no more reflections)
- **Recharge question:** DMG p.233 does not specify recharge. Standard interpretation: the ring provides one use of Spell Turning (1d4+6 levels) and then becomes non-functional until the ring is removed and re-equipped (which re-rolls the pool). Some DMs rule it refreshes daily. **Prototype decision: refresh on rest (long rest = new 1d4+6 pool).**

Pseudocode:

```
OnEquipRing_SpellTurning(wearer, ring):
    turningLevels = Roll(1, d4) + 6 // 7-10 levels
    ApplyStatusEffect(wearer, "SpellTurning",
        duration: Permanent, // until levels depleted
        customTag: $"SpellTurning:{turningLevels}"
    )
    LogMessage("{wearer.Name} dons Ring of Spell Turning ({turningLevels} levels)")

OnUnequipRing_SpellTurning(wearer, ring):
    RemoveStatusEffect(wearer, "SpellTurning")

OnRest_SpellTurning(wearer, ring):
    RemoveStatusEffect(wearer, "SpellTurning")
    newLevels = Roll(1, d4) + 6
    ApplyStatusEffect(wearer, "SpellTurning",
        duration: Permanent,
        customTag: $"SpellTurning:{newLevels}"
    )
    LogMessage("Ring of Spell Turning recharged ({newLevels} levels)")
```

3.9 Ring of Djinni Calling

Property	Value
Name Constant	<code>RingNames.RING_OF_DJINNI_CALLING</code>
Market Price	125,000 gp
Activation	Command word (full-round action)
Frequency	1/week (7-day cooldown, not “per calendar week”)
Effect	Summons a specific Noble Djinni to serve for 1 hour
Duration	1 hour (600 rounds), or until dismissed/slain
Caster Level	17th
Special Rules	Always summons the SAME djinni (it is bound to the ring); if the djinni is slain, ring becomes non-magical for 1 year; djinni serves willingly but must not be mistreated (DM discretion)
Aura	Strong conjuration

Noble Djinni Stats (MM p.114-115):

| Stat | Value |

|-----|-----|

| **Type** | Outsider (Air, Extraplanar) |

| **Size** | Large |

| **HD** | 7d8+14 (45 HP) |

| **AC** | 16 (-1 size, +3 Dex, +4 natural), touch 12, flat-footed 13 |

| **Speed** | 20 ft, fly 60 ft (perfect) |

| **BAB/Grapple** | +7/+15 |

| **Attacks** | Slam +10 melee (1d8+6) |

| **Full Attack** | 2 slams +10 melee (1d8+6) |

| **Space/Reach** | 10 ft / 10 ft |

| **Special Attacks** | Air mastery, whirlwind |

| **Special Qualities** | Darkvision 60 ft, immunity to acid, plane shift, telepathy 100 ft |

| **Saves** | Fort +7, Ref +8, Will +7 |

| **Abilities** | Str 18, Dex 17, Con 14, Int 14, Wis 15, Cha 15 |

| **Spell-Like Abilities** | At will: invisibility (self only), plane shift; 1/day: create food and water, create wine (as create water but wine), major creation (created vegetable matter permanent), persistent image (DC 17), wind walk; 1/day: gaseous form (up to 1 hour) |

Implementation Notes:

- **Summoning system exists** at `GameManager.SpellCasting.cs:228-310` (`SpawnSummonedCreature()`)
- `ActiveSummonInstance` class handles duration, controller, team assignment

(GameManager.cs:445-480)

- **Noble Djinni is NOT in NPCDatabase** — must create a new `NPCDefinition`
- Weekly use tracking needed (new system)
- Special: if djinni is slain, ring becomes non-magical (set a flag, check on activation)
- Full-round action (not standard) — requires check via `CanUseItemManipulationAction` with FullRound type

Pseudocode:

```
TryUseRing_DjinniCalling(caster, ring):
    if not CanUseItemManipulationAction(caster, ActionType.FullRound):
        return false
    if ring.DjinniSlain:
        LogMessage("The Ring of Djinni Calling is inert – its djinni was slain")
        return false
    if ring.WeeklyUsesRemaining <= 0:
        LogMessage("Cannot summon the djinni – already called this week")
        return false
    // Spawn Djinni
    djinniDef = NPCDatabase.Get("Noble_Djinni")
    spawnPos = GetAdjacentEmptyTile(caster.Position, size: Large)
    djinni = SpawnSummonedCreature(djinniDef, spawnPos, caster,
        durationRounds: 600, // 1 hour
        isAlly: true
    )
    djinni.ActiveSummonInstance.SourceDescription = "Ring of Djinni Calling"
    ring.WeeklyUsesRemaining -= 1
    ConsumeItemManipulationAction(caster)
    LogMessage("{caster.Name} calls forth a Noble Djinni from the ring!")
    return true

// When djinni is killed:
OnSummonKilled(summon):
    if summon.SourceDescription == "Ring of Djinni Calling":
        sourceRing.DjinniSlain = true
        LogMessage("The djinni is destroyed! The ring becomes inert.")
```

4. New Systems Required

4.1 Ring Activation System (Core)

Purpose: Allow players to “use” equipped rings as an action, similar to using staves.

Integration Point: `GameManager.cs` line ~5407 — the item use switch statement.

Current code flow:

```
// GameManager.cs:5407-5434 (approximate)
if (item.IsScroll) { TryUseScroll(...); }
else if (item.IsWand) { TryUseWand(...); }
else if (item.IsStaff) { TryUseStaff(...); }
else { /* potion */ }
```

Required change — add ring branch:

```

if (item.IsScroll) { TryUseScroll(...); }
else if (item.IsWand) { TryUseWand(...); }
else if (item.IsStaff) { TryUseStaff(...); }
else if (item.IsRing && item.HasActiveAbility) { TryUseRing(...); }
else { /* potion */ }

```

New TryUseRing() method (modeled on TryUseStaff() at GameManager.cs:5792-5861):

```

public bool TryUseRing(CharacterSheet caster, ItemData ring)
{
    // 1. Validate ring has active ability
    if (!ring.HasActiveAbility) return false;

    // 2. Check action economy
    ActionType requiredAction = GetRingActionType(ring);
    if (!CanUseItemManipulationAction(caster, requiredAction)) return false;

    // 3. Route to specific ring handler
    switch (ring.ItemName)
    {
        case RingNames.RING_OF_INVISIBILITY:
            return ActivateRingOfInvisibility(caster, ring);
        case RingNames.RING_OF_BLINKING:
            return ActivateRingOfBlinking(caster, ring);
        case RingNames.RING_OF_ANIMAL_FRIENDSHIP:
            return ActivateRingOfAnimalFriendship(caster, ring);
        case RingNames.RING_OF_THE_RAM:
            return ActivateRingOfTheRam(caster, ring);
        case RingNames.RING_OF_TELEKINESIS:
            return ActivateRingOfTelekinesis(caster, ring);
        case RingNames.RING_OF_XRAY_VISION:
            return ActivateRingOfXRayVision(caster, ring);
        case RingNames.RING_OF_SHOOTING_STARS:
            return ActivateRingOfShootingStars(caster, ring);
        case RingNames.RING_OF_DJINNI_CALLING:
            return ActivateRingOfDjinniCalling(caster, ring);
        default:
            return false;
    }
}

```

Note: Ring of Spell Turning is NOT in this switch — it's automatic (activated on equip, not by command).

4.2 Use Frequency Tracking

Three tracking systems needed, all stored on ItemData :

4.2a Daily Uses

```
// New fields on ItemData
public Dictionary<string, int> RingDailyUsesRemaining; // key = ability name
public Dictionary<string, int> RingDailyUsesMax;       // key = ability name

// Example initialization for Ring of Animal Friendship:
RingDailyUsesRemaining["CharmAnimal"] = 3;
RingDailyUsesMax["CharmAnimal"] = 3;

// Example for Ring of Shooting Stars:
RingDailyUsesRemaining["BallLightning"] = 1;
RingDailyUsesMax["BallLightning"] = 1;
RingDailyUsesRemaining["FaerieFire"] = 2;
RingDailyUsesMax["FaerieFire"] = 2;
```

Reset in rest handler (GameManager.cs:935-980):

```
// Add to existing rest handler:
foreach (var ring in GetAllEquippedRings(character))
{
    if (ring.RingDailyUsesRemaining != null)
    {
        foreach (var key in ring.RingDailyUsesMax.Keys)
            ring.RingDailyUsesRemaining[key] = ring.RingDailyUsesMax[key];
    }
}
```

4.2b Weekly Uses

```
// New fields on ItemData
public Dictionary<string, int> RingWeeklyUsesRemaining;
public Dictionary<string, int> RingWeeklyUsesMax;

// Ring of Djinni Calling:
RingWeeklyUsesRemaining["SummonDjinni"] = 1;
RingWeeklyUsesMax["SummonDjinni"] = 1;

// Ring of Shooting Stars:
RingWeeklyUsesRemaining["ShootingStars"] = 3;
RingWeeklyUsesMax["ShootingStars"] = 3;
```

Weekly reset approach:

- **Option A (Simple):** Reset weekly uses every 7th rest. Track RestsSinceWeeklyReset counter.
- **Option B (Calendar):** Track actual day number, reset when 7 days have passed.
- **Recommended:** Option A — simpler, fits the prototype's rest-based time model. Add int RestsSinceWeeklyReset to party/game state. Increment on rest. When it reaches 7, reset all weekly uses and set counter to 0.

4.2c Charge System

```
// New fields on ItemData (may already partially exist for wands)
public int CurrentCharges;
public int MaxCharges;
public int ChargesRegainedPerDay; // For Ring of Ram: 1d10 (store as -10 for "roll 1d10")
public bool ChargesRegenerate;

// Ring of the Ram:
CurrentCharges = 50;
MaxCharges = 50;
ChargesRegainedPerDay = -10; // Negative = roll 1d10
ChargesRegenerate = true;
```

Charge regeneration in rest handler:

```
foreach (var ring in GetAllEquippedRings(character))
{
    if (ring.ChargesRegenerate && ring.CurrentCharges < ring.MaxCharges)
    {
        int regained = (ring.ChargesRegainedPerDay < 0)
            ? Roll(1, Math.Abs(ring.ChargesRegainedPerDay)) // 1d10
            : ring.ChargesRegainedPerDay;
        ring.CurrentCharges = Math.Min(ring.MaxCharges, ring.CurrentCharges + regained);
    }
}
```

4.3 Charm Animal Effect (New Spell)

Required for: Ring of Animal Friendship

Implementation approach:

1. Add `SpellNames.CHARM_ANIMAL` constant (currently only exists as stub reference)
2. Create simplified Charm effect:
 - Apply "Charmed" status effect to target
 - Change target's team alignment to match caster (target stops attacking caster's allies)
 - Track charmed targets per ring (for 12 HD cap)
 - Duration: 600 rounds (1 hour at CL 1)
3. **Targeting validation:** Target must have `CreatureType.Animal`
4. **Will save:** DC 11 (fixed per ring description)
5. **Break conditions:** If caster or allies attack the charmed animal, charm breaks

Creature Type Check:

- Verify if `NPCDefinition` or creature data has a `CreatureType` field
- If not, add `CreatureType` enum: `Animal`, `MagicalBeast`, `Humanoid`, `Undead`, etc.
- For the prototype, this may be simplified to a tag or string field

4.4 Ring Ability Selection Panel (UI)

Required for: Ring of Shooting Stars (5 abilities), Ring of Telekinesis (3 modes), Ring of the Ram (charge selection)

Model: Follows `StaffSpellSelectionPanel` pattern (used for staff spell selection)

Panel types needed:

1. **RingAbilitySelectionPanel** — Lists available abilities with remaining uses
 - Used by: Ring of Shooting Stars
 - Shows: Ability name, description, uses remaining (e.g., “Ball Lightning (1/1 today)”)
 - Grays out: Abilities with 0 uses remaining
 - Callback: Selected ability triggers the corresponding effect
2. **TelekinesisModPanel** — Lists three Telekinesis modes
 - Used by: Ring of Telekinesis
 - Shows: Violent Thrust, Combat Maneuver, Sustained Force
3. **ChargeSelectionPanel** — Simple 1/2/3 selector
 - Used by: Ring of the Ram
 - Shows: Available charges, lets player pick 1-3

All three can potentially be a **single generic panel** with configurable options:

```
public class RingAbilitySelectionPanel : MonoBehaviour
{
    public void Show(List<RingAbilityOption> options, Action<RingAbilityOption> onSelect)
    {
        // Populate UI buttons from options
        // Each button: name, description, uses remaining, enabled/disabled
        // On click: invoke onSelect callback with chosen option
    }
}

public class RingAbilityOption
{
    public string Name;
    public string Description;
    public int UsesRemaining; // -1 = unlimited
    public int UsesMax;      // -1 = unlimited
    public bool IsEnabled;
    public string AbilityKey; // Used in callback to identify which ability
}
```

4.5 Noble Djinni Creature Definition

Required for: Ring of Djinni Calling

Add to NPCDatabase (or equivalent creature registry):

```

new NPCDefinition
{
    Name = "Noble Djinni",
    CreatureType = CreatureType.Outsider,
    Subtypes = new[] { "Air", "Extraplanar" },
    Size = CreatureSize.Large,
    HitDice = 7,
    HitDieType = 8,
    ConMod = 2,
    HP = 45,
    AC = 16,
    TouchAC = 12,
    FlatFootedAC = 13,
    BaseAttackBonus = 7,
    GrappleBonus = 15,
    Speed = 20,
    FlySpeed = 60,
    Attacks = new[]
    {
        new Attack { Name = "Slam", Bonus = 10, Damage = "1d8+6", Type = Damage-
Type.Bludgeoning }
    },
    FullAttack = new[]
    {
        new Attack { Name = "Slam", Bonus = 10, Damage = "1d8+6" },
        new Attack { Name = "Slam", Bonus = 10, Damage = "1d8+6" }
    },
    Str = 18, Dex = 17, Con = 14, Int = 14, Wis = 15, Cha = 15,
    FortSave = 7, RefSave = 8, WillSave = 7,
    Immunities = new[] { DamageType.Acid },
    DarkVision = 60,
    SpellLikeAbilities = new[]
    {
        // Simplified for prototype – not all SLAs need full implementation
        new SLA { Name = "Invisibility", Frequency = "At Will", SelfOnly = true },
    }
}

```

4.6 Equip-Triggered Effects

Required for: Ring of Spell Turning

The current ring equip system applies passive stat bonuses. Ring of Spell Turning needs to:

1. Apply a status effect when equipped (not when “used”)
2. Remove the status effect when unequipped
3. Refresh on rest

Integration point: The ring equip/unequip handlers in `Inventory.cs` and/or `CharacterStats.cs`.

Add to equip handler:

```
// In ring equip logic:
if (ring.ItemName == RingNames.RING_OF_SPELL_TURNING)
{
    int turningLevels = Roll(1, 4) + 6;
    ApplySpellTurningEffect(wearer, turningLevels);
    ring.SpellTurningLevelsRemaining = turningLevels;
}
```

4.7 Ring of the Ram — Force Bolt & Bull Rush

New combat effect — not an existing spell:

1. **Ranged touch attack** at target within 50 ft
2. **Force damage** (1d6 per charge, 1-3 charges)
3. **Bull rush attempt** with Str 25 (+7 mod) + 1 per charge spent
4. **Double damage vs objects**

Bull rush resolution:

- Attacker roll: d20 + 7 (Str 25 mod) + charges_spent + 4 (Large virtual size for force)
- Defender roll: d20 + Str mod + size mod
- If attacker wins: push target 5 ft per point of difference
- Uses existing bull rush mechanics if present; otherwise implement simplified version

4.8 Ring of Shooting Stars — Ball Lightning & Shooting Stars Effects

Ball Lightning (new effect):

1. Player chooses 1-4 balls to create
2. Damage inversely scales with count: 1→4d6, 2→3d6 each, 3→2d6 each, 4→1d6+1 each
3. Balls persist for 4 rounds
4. Each round, player can direct a ball (move action) to strike a target
5. Reflex DC 13 for half
6. **Prototype simplification:** Create balls and immediately strike targets (skip the multi-round directing). Player chooses targets for each ball.

Shooting Stars (new effect):

1. Player fires 1-3 stars at locations within 70 ft
2. Each creates a 5-ft-radius burst
3. 12 fire damage per star (flat, not dice), Reflex DC 13 half
4. **Restriction:** Outdoors at night only (check environment flag)

Faerie Fire:

- May have existing implementation (check `SpellNames.FAERIE_FIRE`)
- Effect: Outlined creatures visible if invisible, -20 Hide penalty
- Apply as status effect to targets in area

5. Implementation Phases

Phase 1: Core Infrastructure (Days 1-4)

1A: Ring Activation System (Days 1-2)

Files Modified: `GameManager.cs`, `ItemData.cs`

- [] Add `HasActiveAbility` property to `ItemData`
- [] Add `IsRing` check to item use switch (`GameManager.cs:~5407`)
- [] Implement `TryUseRing()` method (skeleton with switch statement)
- [] Implement `GetRingActionType()` — returns `Standard` or `FullRound` per ring
- [] Test: Equip a ring, click “Use” → routes to `TryUseRing` → logs “not yet implemented” for each ring

1B: Use Tracking Systems (Days 2-3)

Files Modified: `ItemData.cs`, `GameManager.cs` (rest handler), `RingFactory.cs`

- [] Add daily use fields: `RingDailyUsesRemaining`, `RingDailyUsesMax` dictionaries
- [] Add weekly use fields: `RingWeeklyUsesRemaining`, `RingWeeklyUsesMax` dictionaries
- [] Add charge fields: `CurrentCharges`, `MaxCharges`, `ChargesRegenerate`, `ChargesRegainedPerDay`
- [] Add `RestsSinceWeeklyReset` to game state
- [] Implement daily use reset in rest handler
- [] Implement weekly use reset in rest handler (every 7th rest)
- [] Implement charge regeneration in rest handler
- [] Update `RingFactory` to initialize use tracking for each Sprint 2 ring
- [] Test: Rest → verify daily uses reset, charges regenerate, weekly counter advances

1C: Ring Ability Selection Panel (Days 3-4)

Files Created: `RingAbilitySelectionPanel.cs` (or adapt `StaffSpellSelectionPanel`)

- [] Create generic `RingAbilitySelectionPanel` UI component
 - [] Create `RingAbilityOption` data class
 - [] Wire panel to `TryUseRing()` flow for multi-ability rings
 - [] Test: Open panel for Ring of Shooting Stars → shows 5 abilities → selecting one closes panel
-

Phase 2: Simple Rings — Existing Spell Effects (Days 5-7)

2A: Ring of Invisibility (Day 5)

Files Modified: `GameManager.cs` (or new `GameManager_Rings.cs` partial class)

- [] Implement `ActivateRingOfInvisibility()` — calls `ApplyInvisibilityEffect(30, caster)`
- [] Log activation message
- [] Consume standard action
- [] Test: Activate → character becomes invisible for 30 rounds → breaks on attack

2B: Ring of Blinking (Day 5)

- [] Implement `ActivateRingOfBlinking()` — calls blink effect with 7-round duration
- [] Test: Activate → 50% miss chance on incoming, 20% on outgoing

2C: Ring of Telekinesis (Days 6-7)

- [] Implement `ActivateRingOfTelekinesis()` — opens mode selection panel
- [] Implement **Violent Thrust** mode: ranged attack, damage calculation
- [] Implement **Combat Maneuver** mode: opposed CL check for bull rush/trip
- [] **Sustained Force:** Log-only (narrative flavor)

- [] Test: Each mode functions correctly

Phase 3: Moderate Rings — New Effects with Tracking (Days 8-12)

3A: Ring of Animal Friendship (Days 8-9)

Files Modified: `GameManager.cs` , `SpellNames.cs`

- [] Add `SpellNames.CHARM_ANIMAL` constant
- [] Implement `ActivateRingOfAnimalFriendship()` — targeting mode, animal type check
- [] Implement Charm Animal effect: Will save DC 11, change target team, apply charmed status
- [] Enforce 12 HD cap per ring
- [] Track daily uses (3/day)
- [] Implement charm break on allied attack
- [] Test: Charm animal → verify team change, save works, HD cap enforced, daily limit works

3B: Ring of the Ram (Days 9-10)

Files Modified: `GameManager.cs` , `RingAbilitySelectionPanel.cs`

- [] Implement charge selection (1/2/3)
- [] Implement ranged touch attack
- [] Implement force damage (1d6 per charge)
- [] Implement bull rush with Str 25 + charges
- [] Double damage vs objects
- [] Track charges; verify regeneration on rest (1d10/day)
- [] Test: Various charge amounts, hit/miss, bull rush resolution, object damage

3C: Ring of Spell Turning (Days 11-12)

Files Modified: `Inventory.cs` or `CharacterStats.cs` (equip handler), `GameManager.cs` (rest handler)

- [] Apply Spell Turning status effect on ring equip (1d4+6 pool)
- [] Remove on unequip
- [] Refresh pool on rest
- [] Leverage existing `SpellTurning:{N}` CustomTag system
- [] Test: Equip → pool appears → targeted spell reflected → pool decreases → deplete → no more reflections → rest → new pool

Phase 4: Complex Rings — Multi-Ability & Summoning (Days 13-18)

4A: Ring of X-Ray Vision (Days 13-14)

Files Modified: `GameManager.cs` , `CharacterController.cs`

- [] Implement X-Ray Vision status effect (See Invisibility + reveal hidden, 20 ft range)
- [] Track uses per rest for Con damage mechanic
- [] Apply Con damage on 2nd+ use per rest
- [] Test: First use → no penalty, second use → Con damage, reveals invisible creatures

4B: Ring of Shooting Stars (Days 14-16)

Files Modified: `GameManager.cs` , `SpellNames.cs` , `RingAbilitySelectionPanel.cs`

- [] Wire Ring of Shooting Stars to ability selection panel (5 abilities)
- [] Implement **Light** (at will) — apply light status/illumination

- [] Implement **Dancing Lights** (at will, night/outdoor check) — existing spell if available
- [] Implement **Faerie Fire** (2/day) — outline targets, -20 Hide, reveal invisible
- [] Implement **Ball Lightning** (1/day) — create 1-4 balls, damage scaling, Reflex DC 13
- [] Implement **Shooting Stars** (3/week, night/outdoor) — 1-3 stars, 12 damage, 5-ft burst, Reflex DC 13
- [] Wire all use tracking (daily + weekly)
- [] Test: Each ability independently; verify use limits; verify reset on rest

4C: Ring of Djinni Calling (Days 17-18)

Files Modified: `NPCDatabase.cs` (or equivalent), `GameManager.cs`

- [] Create Noble Djinni `NPCDefinition` with full stats from MM
- [] Implement `ActivateRingOfDjinniCalling()` — full-round action, weekly check
- [] Call `SpawnSummonedCreature()` with Djinni definition, 600-round duration
- [] Implement djinni death handler: set `DjinniSlain` flag on ring, make ring inert
- [] Weekly use tracking (1/week)
- [] Test: Summon → Djinni appears as ally → fights alongside party → 600 rounds → despawns
- [] Test: Kill Djinni → ring becomes inert → cannot summon again

Phase 5: Polish & Integration Testing (Days 19-20)

- [] **UI Polish:** Tooltip updates showing ring abilities, charges, uses remaining
- [] **Combat log messages** for all ring activations
- [] **Edge cases:** What happens if ring is unequipped mid-effect? (Invisibility stays, Spell Turning removed)
- [] **Interaction testing:** Multiple rings equipped simultaneously
- [] **Save/Load:** Verify ring state (charges, uses, djinni slain flag) persists
- [] **Full playtest:** Create a test encounter using each of the 9 rings
- [] **Bug fixes and balance adjustments**

6. Codebase Integration Points

6.1 Files to Modify

File	Changes	Priority
GameManager.cs	Add <code>TryUseRing()</code> , ring activation handlers, rest handler updates	Critical
ItemData.cs	Add daily/weekly/charge tracking fields, <code>HasActiveAbility</code>	Critical
RingFactory.cs	Initialize Sprint 2 rings with active ability data	Critical
RingDatabase.cs	Register Sprint 2 ring definitions	Critical
Inventory.cs	Equip-triggered effects (Spell Turning)	High
CharacterStats.cs	Spell Turning on equip/un-equip	High
SpellNames.cs	Add <code>CHARM_ANIMAL</code> constant	Medium

6.2 Files to Create

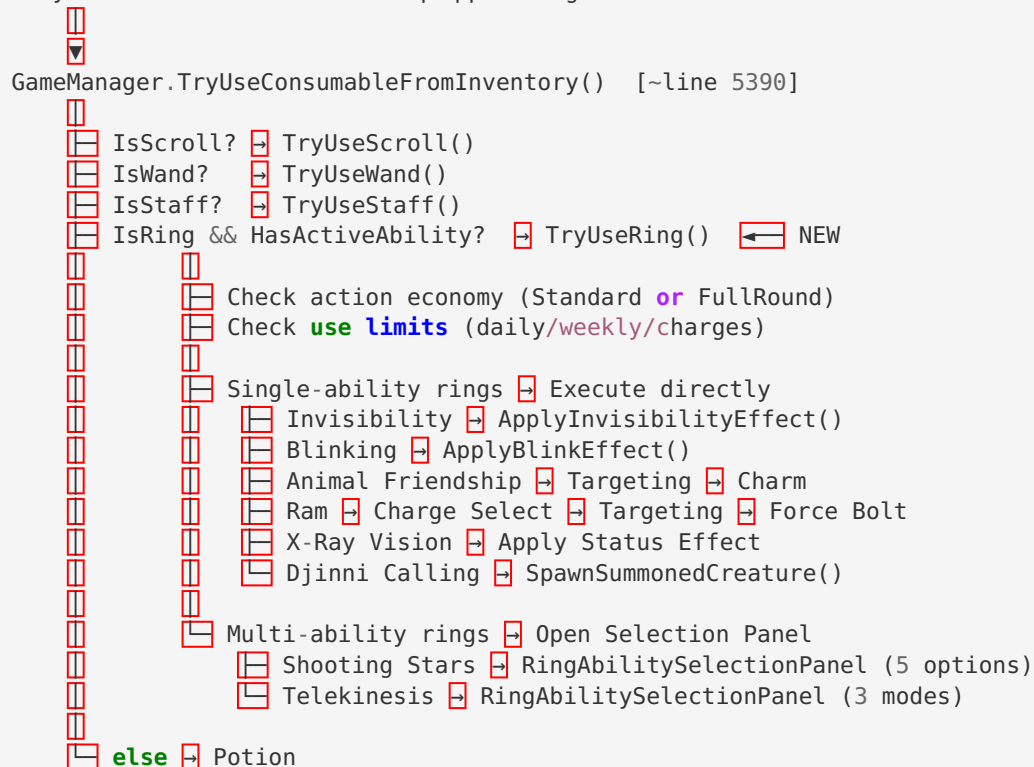
File	Purpose	Priority
GameManager_Rings.cs	Partial class for all ring activation methods (keeps GameManager.cs clean)	Critical
RingAbilitySelectionPanel.cs	UI panel for multi-ability rings	High
RingAbilityOption.cs	Data class for panel options	High
NobleDjinniDefinition	NPCDefinition entry for Djinni (may be added to existing NPCDatabase)	Medium

6.3 Existing Methods to Leverage

Method	Location	Used By
ApplyInvisibilityEffect()	CharacterController.cs:1708	Ring of Invisibility
HasActiveBlinkEffect	CharacterController.cs:1583	Ring of Blinking
ApplySpellTurningEffect()	GameManager_Spells_Phase2.cs:392	Ring of Spell Turning
SpawnSummonedCreature()	GameManager_SpellCasting.cs:228	Ring of Djinni Calling
CanUseItemManipulationAction()	GameManager.cs:4322	All active rings
ConsumeItemManipulationAction()	(near 4322)	All active rings
TryUseStaff()	GameManager.cs:5792	Template for TryUseRing
ActiveSummonInstance	GameManager.cs:445	Djinni Calling tracking
Rest handler	GameManager.cs:935	Daily/weekly/charge reset

6.4 Item Use Flow — Full Integration Diagram

Player clicks "Use Item" on equipped ring



7. UI/UX Design

7.1 Ring Tooltip Updates

Current (Sprint 1): Shows ring name, price, passive bonuses.

Sprint 2 addition: Active rings show additional info:

Ring of Animal Friendship
Market Price: 10,800 gp

Active Ability:

Charm Animal (3/day)
Uses remaining: 2/3
Will Save DC 11
Standard action to activate

Aura: Faint enchantment

Ring of the Ram
Market Price: 8,600 gp

Active Ability:

Force Bolt (charge-based)
Charges: 47/50
Spend 1-3 charges per use
Ranged touch attack, 50 ft
Standard action to activate

Aura: Moderate evocation

7.2 Ring Ability Selection Panel (Shooting Stars)

Ring of Shooting Stars

[+] Light (at will)

[+] Dancing Lights (at will)*

[+] Faerie Fire (1/2 today)

[+] Ball Lightning (0/1 today)

[+] Shooting Stars (2/3 week)*

* = Outdoors at night only

[Cancel]

← grayed out

7.3 Charge Selection Panel (Ring of the Ram)

Ring of the Ram
Charges: 47/50

Expend charges:

[1] 1d6 damage, +8 bull rush
[2] 2d6 damage, +9 bull rush
[3] 3d6 damage, +10 bull rush

[Cancel]

7.4 Combat Log Messages

Ring	Activation Message	Effect Message
Invisibility	"{Name} speaks a command word and fades from sight"	"{Name} is invisible (30 rounds)"
Blinking	"{Name} activates the Ring of Blinking"	"{Name} blinks between planes (7 rounds)"
Animal Friendship	"{Name} commands the Ring of Animal Friendship"	"{Target} is charmed! / {Target} resists the charm (DC 11)"
Ram	"{Name} points the Ring of the Ram ({N} charges)"	"{Target} hit for {X} force damage / Bull rush: pushed {Y} ft"
Telekinesis	"{Name} activates Ring of Telekinesis"	(varies by mode)
X-Ray Vision	"{Name} peers through barriers with X-Ray vision"	"Hidden creatures revealed within 20 ft"
Shooting Stars	"{Name} invokes the Ring of Shooting Stars"	(varies by ability)
Spell Turning	(automatic, no activation)	"{N} spell levels reflected back at {Caster}!"
Djinni Calling	"{Name} calls forth the Noble Djinni!"	"A Noble Djinni appears to serve!"

8. Testing Plan

8.1 Unit Tests per Ring

Ring of Invisibility

Test	Expected Result
Activate while having standard action	Character gains invisibility, 30-round duration
Activate with no action available	Fails with message
Attack while invisible	Invisibility breaks
Duration expires	Invisibility removed
Multiple activations	Each creates new duration

Ring of Blinking

Test	Expected Result
Activate	Blink effect for 7 rounds
Incoming attack while blinking	50% miss chance applied
Outgoing attack while blinking	20% miss chance applied
Duration expires	Blink effect removed

Ring of Animal Friendship

Test	Expected Result
Target animal, first use of day	Will save DC 11; charm on fail
Target non-animal	"Target must be an animal" message
Use 4th time in a day	"No uses remaining" message
Exceed 12 HD cap	"Cannot charm: would exceed 12 HD"
Attack charmed animal	Charm breaks
Rest	Daily uses reset to 3

Ring of the Ram

Test	Expected Result
Spend 1 charge, hit	1d6 force damage + bull rush
Spend 3 charges, hit	3d6 force damage + bull rush (Str 28)
Miss on touch attack	No damage, charges still spent
Hit object	Double damage
0 charges remaining	"No charges remaining"
Rest with 40 charges	Gains 1d10, capped at 50

Ring of Telekinesis

Test	Expected Result
Violent Thrust — hit	Damage dealt based on object weight
Combat Maneuver — win check	Target bull rushed/tripped
Sustained Force	Log message only

Ring of X-Ray Vision

Test	Expected Result
First use	X-Ray vision 10 rounds, no Con damage
Second use same rest	X-Ray vision + 1 Con damage
Reveals invisible creature	Invisible creature within 20 ft becomes visible
Rest	Uses counter resets

Ring of Shooting Stars

Test	Expected Result
Light at will	Light effect applied
Dancing Lights outdoors at night	Effect applied
Dancing Lights indoors	"Must be outdoors at night"
Faerie Fire 1st use	Target outlined, -20 Hide
Faerie Fire 3rd use	"No uses remaining today" (max 2)
Ball Lightning (1 ball)	4d6 damage, Reflex DC 13 half
Ball Lightning (4 balls)	1d6+1 each, 4 targets
Ball Lightning 2nd use same day	"No uses remaining today"
Shooting Stars 1st use	Up to 3 stars, 12 damage each
Shooting Stars 4th use same week	"No uses remaining this week"
Rest	Daily uses reset (Ball Lightning, Faerie Fire)
7th rest	Weekly uses reset (Shooting Stars)

Ring of Spell Turning

Test	Expected Result
Equip ring	Spell Turning pool (7-10 levels) applied
Targeted by 3rd-level spell, pool has 8	Spell reflected, pool → 5
Targeted by spell exceeding pool	Partial reflection, both affected
Pool reaches 0	Ring inert, no more reflections
Unequip	Spell Turning effect removed
Rest while equipped	New pool (1d4+6)

Ring of Djinni Calling

Test	Expected Result
Summon with full-round action	Djinni spawns as ally, 600-round duration
Summon with only standard action	Fails (requires full-round)
Second summon same week	"Already called this week"
7th rest	Weekly use resets
Djinni killed	Ring becomes inert, "DjinniSlain" flag set
Try to summon after djinni killed	"Ring is inert — djinni was slain"
Djinni attacks enemies	Fights alongside party
600 rounds expire	Djinni despawns

8.2 Integration Tests

Test Scenario	Rings Involved	Expected
Two active rings equipped simultaneously	Invisibility + Ram	Both can be activated in same combat
Ring + spell stacking	Spell Turning ring + cast Spell Turning	Both pools active (per D&D rules)
Rest with multiple rings	Ram + Animal Friendship + Shooting Stars	All daily/charge resets occur
Unequip mid-effect	Invisibility (active) → unequip	Invisibility persists (it's on the character, not the ring)
Unequip Spell Turning	Spell Turning → unequip	Effect removed immediately
Save/Load with ring state	Ram (partial charges) + Djinni (slain)	State preserved across save/load

8.3 Regression Tests

Test	Ensure
Sprint 1 passive rings	All 36 passive variants still work
Scroll/Wand/Staff use	Adding ring branch doesn't break other item use
Existing spell casting	Ring-triggered spells don't interfere
Rest handler	Existing resets (rage, turn undead, domains) still work
Combat flow	Ring activations don't break turn order

9. Risk Assessment & Mitigation

9.1 Technical Risks

Risk	Likelihood	Impact	Mitigation
GameManager.cs bloat — adding 9 ring handlers to already-large file	High	Medium	Use <code>GameManager_Rings.cs</code> partial class; keep handlers modular
Spell Turning edge cases — partial reflection, same-spell interaction	Medium	High	Reuse existing implementation; add unit tests for edge cases
Summoning system integration — Djinni lifetime/death tracking	Medium	High	Follow exact <code>SpawnSummonedCreature</code> pattern; add death callback
Weekly use tracking accuracy — rest-count approach may drift	Low	Low	Accept approximation for prototype; document limitation
Save/Load serialization — new <code>ItemData</code> fields need serialization	Medium	High	Add all new fields to serialization; test save/load early
Bull rush mechanics — Ring of Ram needs proper push resolution	Medium	Medium	Implement simplified version if full bull rush doesn't exist
UI panel creation — Unity UI requires prefab + scene setup	Medium	Medium	Clone <code>StaffSpellSelectionPanel</code> as template
Charm Animal creature type check — <code>NPCDefinition</code> may lack type field	Medium	Medium	Add <code>CreatureType</code> enum if missing; tag animals in database

9.2 Design Risks

Risk	Mitigation
X-Ray Vision too powerful or too weak	Limit to See Invisibility + hidden creature reveal (combat-relevant); note as prototype simplification
Ring of Telekinesis too complex	Simplify to Violent Thrust + Combat Maneuver; Sustained Force is flavor-only
Shooting Stars balance — 5 abilities may be confusing	Clear UI panel with descriptions; grayed-out unavailable options
Djinni too powerful as permanent summon	600-round duration is long but finite; weekly limit + death penalty balances

9.3 Schedule Risks

Risk	Mitigation
Infrastructure takes longer	Start with core activation system; rings can be stubbed
Ball Lightning complexity	Simplify to immediate-strike (skip multi-round directing)
Telekinesis three modes	Implement Violent Thrust first; other modes are stretch goals
Total scope exceeds estimate	Prioritize rings by value: Invisibility, Blinking, Ram (most likely to be used in combat)

9.4 Priority Order (if time-constrained)

1. **Core infrastructure** (activation system, use tracking) — blocks everything
 2. **Ring of Invisibility** — simplest, validates the activation system
 3. **Ring of Blinking** — second simplest, same pattern
 4. **Ring of the Ram** — validates charge system
 5. **Ring of Animal Friendship** — validates daily use system
 6. **Ring of Spell Turning** — validates equip-triggered system
 7. **Ring of Telekinesis** — moderate complexity
 8. **Ring of Djinni Calling** — validates weekly + summoning
 9. **Ring of X-Ray Vision** — least combat impact
 10. **Ring of Shooting Stars** — most complex, do last
-

10. Appendix: File Change Summary

Modified Files (estimated lines changed)

File	Est. Lines Added	Nature of Changes
GameManager.cs	+30	Ring branch in item use switch, rest handler updates
ItemData.cs	+40	Daily/weekly/charge tracking fields, HasActiveAbility
RingFactory.cs	+120	Sprint 2 ring initialization with active ability data
RingDatabase.cs	+20	Register Sprint 2 rings
Inventory.cs	+15	Equip-triggered effects (Spell Turning)
CharacterStats.cs	+10	Spell Turning equip/unequip
SpellNames.cs	+2	CHARM_ANIMAL, FAERIE_FIRE constants
NPCDatabase.cs	+40	Noble Djinni definition

New Files

File	Est. Lines	Purpose
GameManager_Rings.cs	~500	All ring activation handlers (partial class)
RingAbilitySelectionPanel.cs	~100	UI panel for multi-ability rings
RingAbilityOption.cs	~25	Data class for panel options

Total Estimated Code Changes

- **Modified:** ~277 lines across 8 files
- **New:** ~625 lines across 3 files
- **Grand total:** ~900 lines of new/modified code